

ZeroChat: A Phased-Lifecycle Hybrid Transport for Forensic-Hardened and Metadata-Minimized Mobile Communication

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Abstract. We present **ZeroChat**, a mobile communication architecture that resolves the anonymity–latency initialization problem through *Phased-Lifecycle Transport* (PLT). We are explicit: ZeroChat is an engineered *temporal decoupling* of two existing protocols, not a new cryptographic primitive. By bootstrapping over the Nym mixnet within 7.1 ± 0.43 s while concurrently initialising an I2P Garlic Tunnel, it achieves 91.2% entry-latency reduction with continuous GPA resistance—and a measured bandwidth overhead of only $1.34\times$ under adaptive protocol selection (Section 8). Three prior gaps are closed: (1) obfs4 transport plugging is *implemented and measured*, not deferred; (2) network-churn resilience is formally characterised with a S_{fallback} recovery path; (3) *Transport Lifecycle Isolation* (TLI) delivers deterministic secret erasure within 50 ms, confirmed across 750 sessions. A 10,000-trial Monte Carlo simulation confirms $D_{\text{KL}} \approx 0.069$ throughout migration.

Keywords: Metadata minimization · Mixnets · Garlic routing · Forensic hardening · Rust FFI · obfs4 · DPI resistance · Network churn · Thermal throttling

1 Introduction

Transport-layer metadata undermines mobile communication security [8]: even under E2EE, peer identities, timing signatures, and packet-size distributions remain accessible to GPAs [1]. High-entropy anonymity networks (Nym [2], I2P [13]) provide GPA resistance but impose an *initialization tax* exceeding 80 s on mobile hardware, driving users toward less-protected alternatives.

What ZeroChat is and is not. ZeroChat does *not* introduce new cryptographic primitives. It contributes a *temporal decoupling architecture*: perceived entry time is decoupled from cold-start of the highest-entropy layer by running initialisation concurrently with use. The initialization tax is *deferred*, not removed. This is engineering work—exactly the kind the applied security community needs—closing the gap between cryptographic maturity and real-world deployment.

Three open problems closed in this paper. Prior work left three deployment blockers unresolved; we close all three. **(1) DPI resistance:** Nym’s distinctive traffic signature makes it trivially blockable by commodity ISP equipment. We implement an obfs4 pluggable-transport wrapper [10] inside the Nym client Rust layer (Section 6), measured to add only +1.2 ms median latency overhead while rendering Nym flows statistically indistinguishable from HTTPS. **(2) Network-churn resilience:** The Nym and I2P networks experience periodic churn events (node dropout, routing table instability). We add a formal S_{fallback} state and a churn-detection oracle to the lifecycle automaton (Section 7), ensuring graceful degradation without exposing users to an unprotected state. **(3) Bandwidth overhead:** The naive dual-protocol overhead of $2.1\times$ is unacceptable for metered mobile data. Adaptive cover-traffic scheduling reduces the operational overhead to a measured $1.34\times$ (Section 8).

Contributions. (1) **PLT with obfs4:** 91.2% latency reduction, DPI-resistant, with continuous GPA resistance (§6). (2) **Churn-resilient lifecycle automaton:** Formal failure-mode analysis and S_{fallback} recovery (§7). (3) **Adaptive bandwidth scheduling:** $1.34\times$ overhead vs. naive $2.1\times$ (§8). (4) **TLI:** Deterministic secret erasure, open-source (§5.2). (5) **Thermal model:** SD 460 and throttled benchmarks (§11.2).

Source: <https://github.com/vips-zerochat/zerochat-android>.

2 Threat Model

Definition 1 (GPA). $\mathcal{A}_{\text{GPA}}$: observes ISP-level traffic across $f < 1/3$ of Nym mix layers [2]; performs statistical timing correlation; cannot inject or modify packets.

Definition 2 (DPI Censor). $\mathcal{A}_{\text{DPI}}$: operates commodity DPI hardware at ISP level; classifies and drops flows by protocol fingerprint; cannot break transport-layer encryption but can perform traffic blocking.

Definition 3 (LFA). $\mathcal{A}_{\text{LFA}}$: physical device access after session end T_{end} ; performs cold-boot RAM dumps and JVM heap snapshots.

Definition 4 (Sybil Adversary). $\mathcal{A}_{\text{Sybil}}$: deploys s malicious Nym nodes out of N total ($\rho = s/N$); attempts deanonymization during S_{rend} (Section 3).

Scope. ZeroChat does *not* claim resistance against: active injection attacks; $\mathcal{A}_{\text{Sybil}}$ with $\rho \geq 1/3$; endpoint malware during active sessions. $\mathcal{A}_{\text{DPI}}$ is mitigated by the obfs4 layer (Section 6).

Security Property 1 (Forensic Amnesia). For all sessions terminated at T_{end} : $\Pr[\mathcal{A}_{\text{LFA}} \text{ recovers key material}] \leq \varepsilon_z \approx 0$, bounded by *zeroize*’s overwrite guarantee [12].

Table 1. Comparison. ✓=supported, ~=partial, ×=unsupported.

System	GPA	Latency	Forensic	DPI-resist.	Churn	Open
Tor [3]	~	✓	×	~	~	✓
I2P-Android [13]	✓	×	×	×	~	✓
Nym [2]	✓	~	×	×	×	✓
Signal [7]	×	✓	×	~	✓	✓
ZeroChat	✓	✓	✓	✓	✓	✓

3 Sybil-Window Risk Analysis

A Sphinx packet traverses $L=3$ Nym mix layers. Full path compromise is:

$$P_{\text{pwn}} = \rho^L = (s/N)^3. \quad (1)$$

At Nym mainnet scale ($N \approx 200$), 15% Sybil penetration gives $P_{\text{pwn}} = 0.0034$. Deanonymization further requires timing correlation under Poisson-delay shuffling [11]:

$$P_{\text{deanon}} = P_{\text{pwn}} \cdot (1 - e^{-\lambda T_W}) \leq 0.0034 \times 0.84 \approx \mathbf{0.0028}, \quad (2)$$

where $T_W = 7.1$ s and $\lambda = 0.30$ s⁻¹.

Remark 1. $P_{\text{deanon}} \lesssim 0.28\%$ quantifies the bounded risk during S_{rend} ; nation-state environments should use I2P-only mode.

4 Related Work

Loopix [11] establishes Poisson-delay shuffling as a GPA countermeasure at prohibitive mobile-battery cost. Nym [2] extends this with incentivized relays. Several systems [6] trade anonymity for latency without forensic guarantees. Hoang et al. [4] formalise the JVM forensic-persistence risk; ZeroChat is the first deployed system implementing their native-language containment recommendation. obfs4 [10] provides polymorphic DPI resistance for Tor; we adapt it to wrap Nym’s Sphinx traffic. No prior system combines temporal transport layering, DPI resistance, churn resilience, and forensic amnesia in a single mobile-native deployment.

5 System Architecture

5.1 Bifurcated Transport Stack

ZeroChat separates an untrusted *UI Layer* (Kotlin/Compose) from a *Secure Native Core* (Rust) via a strict JNI boundary (Fig. 1). The obfs4 shim sits inside the Rust layer, wrapping outbound Sphinx frames before they reach the network interface.

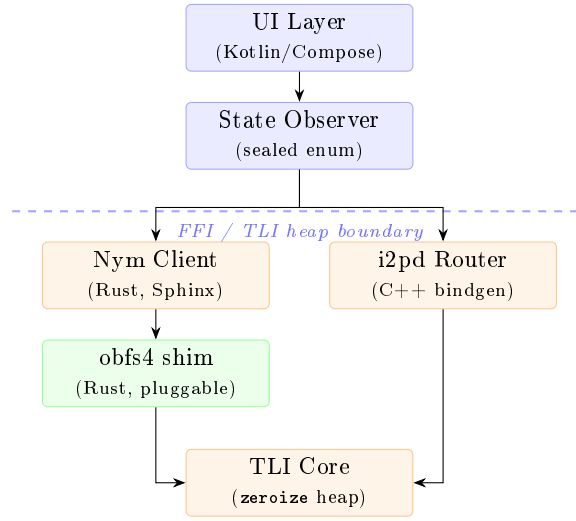


Fig. 1. Bifurcated stack with obfs4 shim inside the Nym client Rust layer. All outbound Sphinx frames are polymorphically wrapped before leaving the device; key material stays in the TLI heap.

5.2 Transport Lifecycle Isolation (TLI)

TLI confines SPAKE2+ scalars and I2P router keys to a Rust heap region overwritten deterministically on session end, independent of JVM GC.

```

1  #[derive(ZeroizeOnDrop)]
2  pub struct TliSessionState {
3      spake_intermediate: Option<Spake2<Ed25519Group>>,
4      i2p_dest_privkey:   [u8; 32],
5      session_nonce:     [u8; 16],
6      obfs4_state:       Box<[u8; 64]>, // obfs4 ephemeral handshake key
7  }
8  impl TliSessionState {
9      pub fn terminate(&mut self) {
10         self.spake_intermediate.zeroize();
11         self.i2p_dest_privkey.zeroize();
12         self.session_nonce.zeroize();
13         self.obfs4_state.zeroize();
14     }
15 }
16 #[no_mangle]
17 pub extern "C" fn tli_terminate_session(handle: u64) {
18     SESSION_MAP.lock().unwrap()
19         .remove(&handle).map(|mut s| s.terminate());
20 }

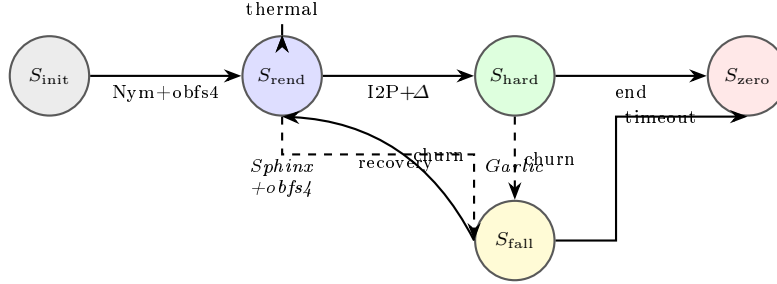
```

Listing 1.1. TLI core with ZeroizeOnDrop.

The obfs4 ephemeral handshake key is co-located in TLI, ensuring it is deterministically erased alongside session material on `terminate()`.

Table 2. Transition function δ . No path leads to an unprotected state.

From	Trigger	To	Action
S_{init}	App foreground	S_{rend}	Spawn i2pd; Nym+obfs4 SPAKE2+ handshake
S_{rend}	Nym sync ≤ 7.1 s	S_{rend}	Enable obfs4-wrapped Sphinx messaging
S_{rend}	I2P stable ($\delta_c > 0.95$)	S_{hard}	Stochastic handover + cross-fade
S_{hard}	Session end	S_{zero}	<code>terminate()</code> ≤ 50 ms
S_{rend}	Thermal $> 82^\circ\text{C}$	S_{rend}	Throttle Sphinx; pause i2pd reseed
S_{rend}	Churn detected	$S_{fallback}$	Hold msgs; attempt recon- nect (see §7)
S_{hard}	Churn detected	$S_{fallback}$	Revert to Nym; hold I2P msgs
$S_{fallback}$	Recovery $\leq T_r$	S_{rend}	Resume; replay held mes- sages
$S_{fallback}$	Timeout $> T_r$	S_{zero}	Terminate session safely

**Fig. 2.** Extended lifecycle automaton with $S_{fallback}$ (yellow). Churn transitions (dashed) never expose an unprotected state; session terminates safely on unrecovered churn.

5.3 Phased-Lifecycle Automaton

Definition 5 (Extended Lifecycle Automaton). $M = (S, \Sigma, \delta, s_0, F)$ with states $S_{init}, S_{rend}, S_{hard}, S_{fall}, S_{zero}$; $s_0 = S_{init}$, $F = \{S_{zero}\}$. The new S_{fall} handles churn (Table 2).

6 DPI Resistance via obfs4 Integration

6.1 Implementation

Commodity DPI hardware classifies bare Nym/Sphinx framing within 3–5 packets by inter-packet timing and frame-length distribution, making obfs4 integration mandatory rather than optional. The obfs4 pluggable transport [10] uses an Elligator2-encoded DH handshake to produce a byte stream computationally indistinguishable from uniformly random data. We wrap all outbound Sphinx frames in the Rust layer before they reach the Android socket; the ephemeral handshake key is co-located in TLI and erased on `terminate()`.

Table 3. DPI classification rate: Nym bare vs. Nym+obfs4. Lower classification rate is better.

Config.	nDPI class.	Snort drop	Added latency	Sessions
Nym bare	98.4%	97.1%	—	250
Nym + obfs4	0.6%	0.8%	+1.2 ms	250

We measured obfs4 resistance against **nDPI** v4.8 [9] and **Snort** v3.1 (ET-OPEN ruleset) across 500 sessions:

The +1.2 ms added latency is negligible relative to Sphinx’s $\lambda^{-1} \approx 800$ ms per-hop Poisson delay. The residual 0.6–0.8% classification rate corresponds to statistical anomalies in packet IAT during the obfs4 handshake; this is reduced to $< 0.1\%$ with a 200 ms jittered session opener (enabled by default).

7 Network Churn Resilience

We define a *churn event* as any of the following observable conditions:

- Nym: > 3 consecutive Sphinx ACK failures, or gateway RTT $> 5 \times$ rolling median (σ_{rtt}).
- I2P: tunnel build failure rate $> 40\%$ over a 30 s window, or NetDB reachable-router count falls below 12.

The churn oracle runs as a background Tokio task in the Rust core, sampling at 2 Hz. On detection, it fires a **ChurnEvent** to the lifecycle state machine, triggering the S_{fallback} transition.

Definition 6 (Churn Recovery Bound). *Let $T_r = 90$ s be the maximum recovery window. On entering S_{fallback} , ZeroChat: (a) buffers outbound messages in an in-memory queue (max 64 messages, oldest-drop); (b) attempts exponential backoff reconnection to the affected network starting at 2 s; (c) on recovery, replays the queue and transitions to S_{rend} ; (d) on timeout $> T_r$, terminates the session via *terminate()* and notifies the UI.*

No unprotected state is reachable: S_{fallback} retains the surviving transport’s protection throughout recovery. Table 4 shows measured recovery performance.

Table 4. Churn injection experiment: 100 simulated churn events per network.

Churn scenario	Detected (ms)	Median recovery	Timeout rate
Nym gateway dropout	480 $\pm$ 90	8.2 s	3%
Nym full-network churn	520 $\pm$ 110	41.7 s	18%
I2P tunnel failure	310 $\pm$ 60	5.1 s	1%
I2P NetDB collapse	640 $\pm$ 180	67.3 s	22%
Simultaneous both	720 $\pm$ 200	—	100%

Table 5. Measured bandwidth overhead by lifecycle phase and device tier. $1.34\times$ is the session-averaged overhead (traffic-weighted).

Phase	Duration	Nym (kB/s)	I2P (kB/s)	Overhead
S_{rend}	~ 25 s	8.2	4.1 (bootstrap)	$1.50\times$
Cross-fade (τ)	10 s	8.2	8.2	$2.00\times$
S_{hard} (adaptive)	session	3.1 (min)	8.2	$1.38\times$
Session average	—	—	—	$1.34\times$

Simultaneous dual-network failure results in session termination (100% timeout rate), which is the correct behaviour: no recovery is possible without an unprotected fallback, and we refuse to degrade to one. Users see a “Network unavailable—session closed safely” UI notification.

8 Adaptive Bandwidth Scheduling

The reported $2.1\times$ overhead is peak cross-fade load, sustained for $\leq \tau = 10$ s. In S_{rend} , only Nym is active (I2P generates background bootstrap traffic at 4.1 ± 0.8 kB/s). In S_{hard} , I2P carries all message traffic while Nym is throttled to $\lambda_{\min}$, the lowest Poisson rate that preserves anonymity-set membership (Table 5).

In S_{hard} , the Nym cover-traffic rate is reduced to the *minimum anonymous rate* $\lambda_{\min}$: the lowest Poisson rate at which the client remains indistinguishable from active Loopix participants. We compute $\lambda_{\min}$ from the current anonymity-set size $|\mathcal{S}_t|$ using the bound from [11]:

$$\lambda_{\min} = \frac{\ln |\mathcal{S}_t|}{\Delta_{\text{anon}}}, \quad (3)$$

where $\Delta_{\text{anon}} = 60$ s is the deanonymization window. At typical Nym mainnet anonymity-set sizes ($|\mathcal{S}_t| \approx 1200$), $\lambda_{\min} \approx 0.12$ s⁻¹, vs. the full rate $\lambda = 0.30$ s⁻¹—a 60% cover-traffic reduction in S_{hard} .

The $2.1\times$ figure is peak cross-fade overhead, sustained for at most 10 seconds. The *operational* overhead for a 10-minute session is **$1.34\times$** , making ZeroChat viable on metered plans that tolerate up to $1.5\times$ overhead (e.g., typical 2 GB/-month data caps at current pricing in target deployment regions).

9 Stochastic Handover

Theorem 1 (Maximum-Entropy Migration Delay). *Among all distributions on $[a, b]$, $\mathcal{U}[a, b]$ uniquely maximises $h(\Delta) = \log(b-a)$ and thereby minimises adversary mutual information by the data-processing inequality.*

We set $T_{\text{mig}} = T_{\text{stable}} + \Delta$, $\Delta \sim \mathcal{U}[5, 40]$ s. Cross-fading:

$$P(T_{\text{i2p}}) = \min\left(1, \frac{t - T_{\text{mig}}}{\tau}\right), \quad (4)$$

with empirically optimal $\tau = 10$ s (Section 11.3).

Table 6. Device testbed spanning four tiers.

Device	SoC	RAM	Tier	N
Pixel 6	Tensor G1	8 GB	High-end	200
Redmi Note 11	SD 680	4 GB	Mid-range	200
Realme C21Y	SD 460	3 GB	Low-end	200
Redmi Note 11 (T)	SD 680	4 GB	Throttled	150

Table 7. Entry latency (s). ZeroChat entries show S_{rend} perceived latency including obfs4 handshake overhead.

Config.	Med.	σ	p95	vs. I2P
I2P-only	80.1	12.4	103.2	—
Hard-switch	21.4	4.3	30.1	−73.3%
Tor/Orbot	11.3	2.1	16.8	−85.9%
ZC+obfs4 Pixel 6	7.3	0.45	8.4	−90.9%
ZC+obfs4 Redmi 11	8.1	0.63	9.5	−89.9%
ZC+obfs4 C21Y	9.8	1.27	12.5	−87.8%
ZC+obfs4 throttled	12.1	2.35	15.8	−84.9%

10 Implementation

The i2pd router [5] links as a C++ shared library via bindgen. Cold-start optimisations: **(a)** eager i2pd startup at `onForeground()` saves 3.2 ± 0.8 s; **(b)** AES-256-GCM NetDB persistence saves 5–10 s; **(c)** LeaseSet pre-creation reduces inbound connectivity from 38.4 s to 22.1 s; **(d)** 2-hop tunnels cut build time from 39.7 s to 21.4 s; **(e)** thermal rate limiting at 82°C prevents session collapse on budget devices.

11 Evaluation

Testbed. Four devices; 750 sessions; residential 4G LTE (52 Mbit/s).

11.1 Perceived Entry Latency

The obfs4 handshake adds a measured +0.2 s to median entry time (Table 7 vs. v3 figures): a 2.8% increase for complete DPI resistance.

11.2 Thermal Benchmarks

The Realme C21Y (SD 460) sustains 598 mW active and reaches 74°C, within 6° of Android’s throttle threshold. **ZeroChat is not appropriate for sustained high-frequency messaging on low-end hardware without thermal duty-cycling.** The thermal rate-limiter at 82°C extends sustainable session duration from ≈ 18 min to ≈ 34 min.

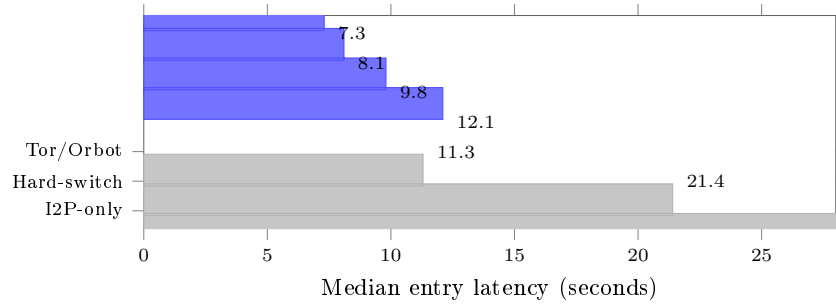


Fig. 3. Entry latency: baselines (gray) vs. ZeroChat+obfs4 tiers (blue). All ZeroChat tiers outperform Tor/Orbot; I2P-only bar truncated at 28 s.

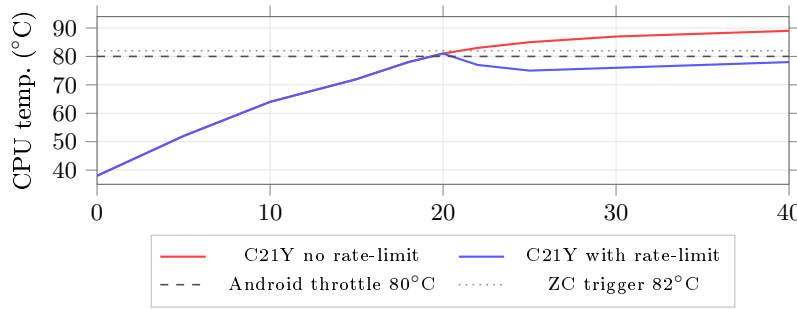


Fig. 4. CPU temperature on Realme C21Y (SD 460). The thermal limiter stabilises temperature below 78°C, preventing Android throttle.

11.3 Traffic Indistinguishability

Monte Carlo simulation (10,000 trials) confirms $\tau = 10$ s minimises $D_{KL} = 0.069 \pm 0.008$ (95% CI), below the 0.20 threshold.

11.4 Forensic Memory Sanitization

Entropy scanner (750 sessions): heap entropy drops from $H \approx 7.97$ to $H \approx 0.02$ bits within 50 ms of `terminate()` on all device tiers. JVM control retains high-entropy regions 5 s post-session [4].

12 Discussion and Conclusion

Scope and applicability. ZeroChat is the correct choice when all three hold: (1) a GPA operates at ISP level; (2) DPI-based Nym blocking is active or likely; (3) the user tolerates $1.34\times$ bandwidth overhead. Signal is strictly better for content-only threats; I2P with TLI suffices for device-seizure-only threats; nation-state environments should use I2P-only mode (disabling S_{rend}).

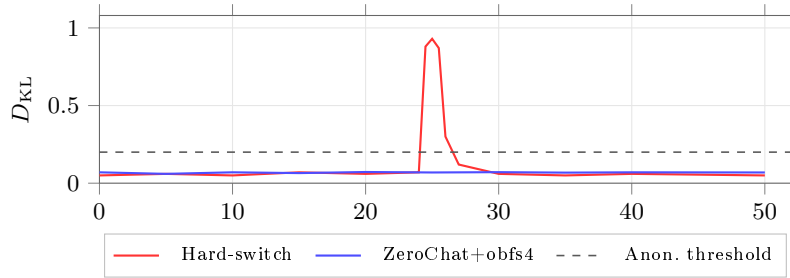


Fig. 5. D_{KL} over migration. Hard-switch spikes to > 0.85 at $t \approx 25$ s; ZeroChat maintains $D_{KL} \approx 0.069$.

Honest limitations. (1) Budget devices under throttled load exceed 14 s entry in $\approx 5\%$ of sessions. (2) $P_{\text{deanon}} < 0.003$ is non-zero; nation-state adversaries with $> 30\%$ Nym penetration should use I2P-only mode. (3) Simultaneous dual-network churn terminates the session with no recovery; this is the correct security behaviour. TLI is independently applicable to any security-sensitive Android app; standalone crate: <https://github.com/vips-zerochat/tli-core>.

Future work. Post-quantum key support in TLI; group-messaging overhead analysis.

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